Generating the Past, Present and Future from a Motion-Blurred Image Supplemental Material

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CCS Concepts: • **Computing methodologies → Computer vision**; **Computational photography**.

Additional Key Words and Phrases: exposure control, deblurring, video diffusion model

ACM Reference Format:
SaiKiran Tedla, Kelly Zhu, Trevor Canham, Felix Taubner, Michael S. Brown, Kiriakos N. Kutulakos, and David B. Lindell. 2025. Generating the Past, Present and Future from a Motion-Blurred Image Supplemental Material. *ACM Trans. Graph.* 44, 6, Article 202 (December 2025), 3 pages. https://doi.org/10.1145/3763306

S1 Supplemental Results

S1.1 Webpage

Please refer to the supplemental webpage for video results: **webpage-supplemental/index.html**. We include video results and comparisons to baselines for in-the-wild generation of the past, present and future from a motion-blurred image; additional video results on bringing historical images to life; 4D scene reconstruction; 3D human pose estimation from our output video sequences; multi-modality of generated videos; and ablations on exposure interval control.

S1.2 Additional 3D Consistency Results

We evaluate the 3D consistency of the model output by analyzing a video from a scene with motion blur due to viewpoint movement. We apply SIFT [Lowe 2004] and RANSAC [Fischler and Bolles 1981] to the first and last generated frames to detect feature correspondences and compute the fundamental matrix [Hartley and Zisserman 2003]. Additionally, we utilize RANSAC [Fischler and Bolles 1981] to find the 2D homography that best explains the keypoint correspondences between both images. We visualize the epipolar lines and the homography applied to both the keypoints and images in Figures S1. While the homography aligns well with the background, it produces errors on the foreground bushes closer to the camera, revealing parallax effects from the scene's 3D structure.

S2 Supplemental Implementation Details

S2.1 Model

CogVideoX. The proposed method is a finetuned version of CogVideoX [Yang et al. 2025], an open-source video diffusion transformer. The model combines a 3D causal variational autoencoder (VAE) and a transformer network. The VAE takes in a sequence of frames and performs spatio-temporal compression to encode them into latent frames. More specifically, the input video frames pass through two layers of spatio-temporal down-sampling and one additional layer of spatial down-sampling with interleaved ResNet encoder stages.

The latent encoding achieves a compression rate of $4 \times 8 \times 8$ for a video of dimensions (F, H, W). However, note that the temporal compression rate is only approximately four—the VAE temporal latent encoding converts the first video frame into its own latent frame, and then every group of four subsequent frames is compressed into a latent frame. In our setup, we convert the input blurry image into one latent frame and predict 16 individual frames which results in 5 latent frames. Thus, our diffusion model gets as input patches from 6 latent frames. We note that in cases where we predict less than 16 frames, we replicate the last frame of the sequence and give it the same exposure interval encoding. This padding is done at both training time and sampling time.

Position encoding. We take advantage of the existing CogVideoX [Yang et al. 2025] sinusoidal positional embedding that encodes the spatio-temporal location of all the patches input to the transformer. They apply a 1D sinusoidal encoding γ_t to the temporal coordinate t, and a 2D sinusoidal encoding γ_s to the spatial coordinates (x, y). The final position encoding is given by:

$$\gamma(x, y, t) = \textsc{Concat}(\gamma_t(t), \gamma_s(x, y)) \in \mathbb{R}^D , \quad \text{(S1)}$$

where $\gamma_t : \mathbb{R} \to \mathbb{R}^{D_t}$ and $\gamma_s : \mathbb{R}^2 \to \mathbb{R}^{D_s}$ denote sinusoidal encoding functions for time and space, respectively. Each function maps its

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ACM 1557-7368/2025/12-ART202
https://doi.org/10.1145/3763306

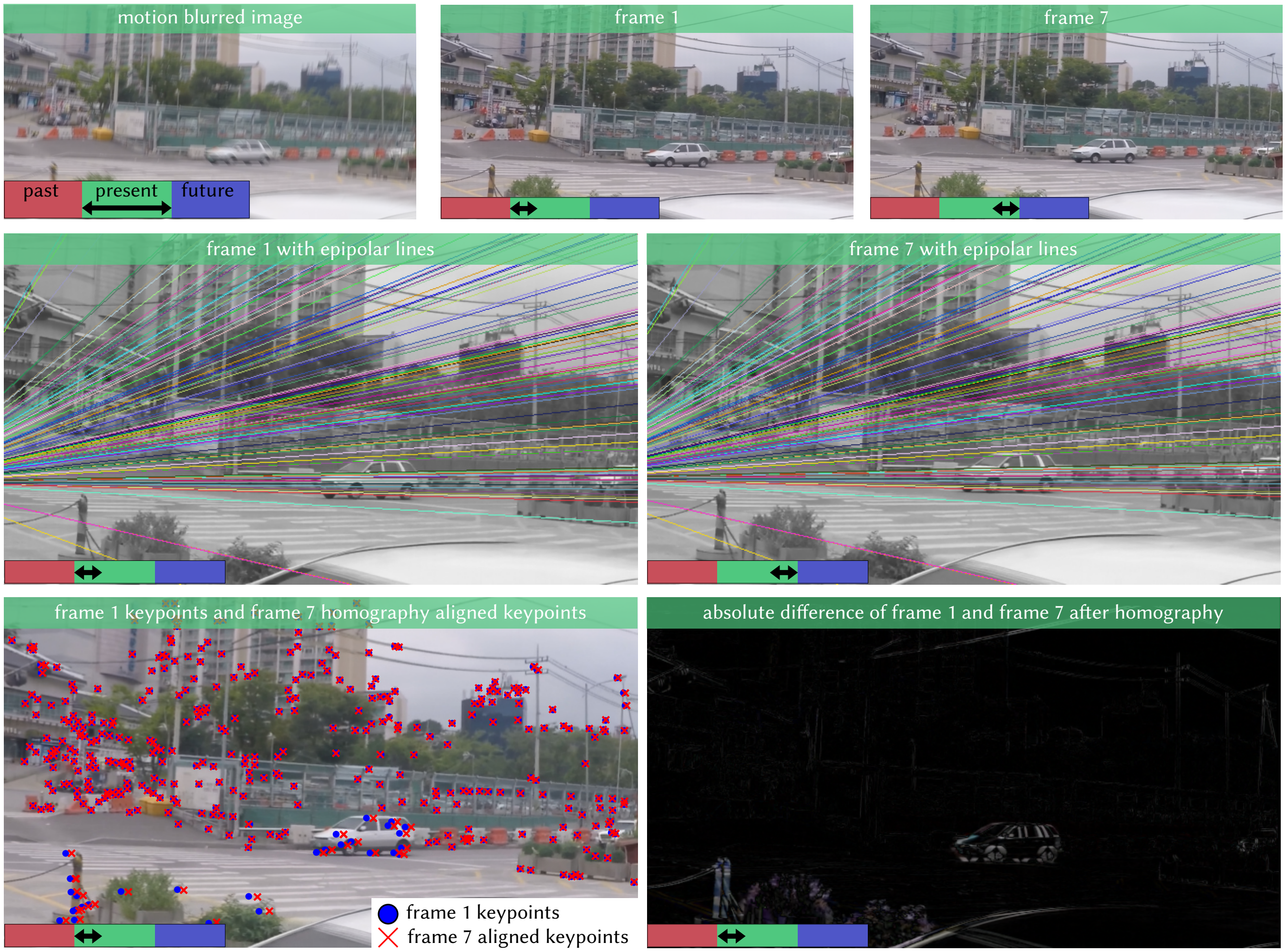

Fig. S1. We assess 3D consistency of an image from the GoPro dataset [Nah et al. 2017] with motion blur due to translation. We visualize the epipolar lines, the residual movement of keypoints after applying a 2D homography that best aligns the two frames, and the absolute difference of the first frame with the homography-warped version of the last frame. We find the epipolar lines to be consistent with the 3D translation (approximately) parallel to the image plane. Additionally, we observe stronger parallax compared to the more distant trees and buildings.

inputs using fixed frequency bands:

$$\gamma_t(t) = [\cos(2\pi\nu_1 t), \sin(2\pi\nu_1 t), \ldots, \cos(2\pi\nu_{D_t/2} t), \sin(2\pi\nu_{D_t/2} t)], \tag{S2}$$

$$\gamma_s(x, y) = \textsc{Concat}\,(\gamma_1(x), \gamma_2(y)) , \tag{S3}$$

where γ_1 and γ_2 are 1D encodings applied to x and y respectively, each with $D_s/2$ dimensions. The full embedding $\gamma(x, y, t)$ is tiled across the spatial and temporal grid and added to the input latent tokens before applying the transformer attention layers.

Since we pass the blurred image latent as the first latent frame, it receives the embeddings corresponding to frame 1. Thus, the model is able to always reliably differentiate the input blurred frame from the frames it is generating.

Exposure interval encoding. In order to condition the network to return video frames with arbitrary exposure start times and durations, we define an exposure interval encoding. Based on the temporal compression ratio, the i-th frame of the latent video is associated with four exposure intervals, which we represent as a vector $\tilde{\mathcal{T}}_i \in \mathbb{R}^8$. For the first latent frame, for example, $\tilde{\mathcal{T}}_1 = [\mathcal{T}_1[1], \mathcal{T}_1[2], \ldots, \mathcal{T}_4[1], \mathcal{T}_4[2]]$.

In addition to these positional embeddings, we also apply an exposure interval encoding to indicate the start and end times of each latent frame in the normalized interval $\mathcal{T} = [-0.5, 0.5]$ such that the exposure intervals $\mathcal{T}_1..\mathcal{T}_4$ are expressed relative to this interval.

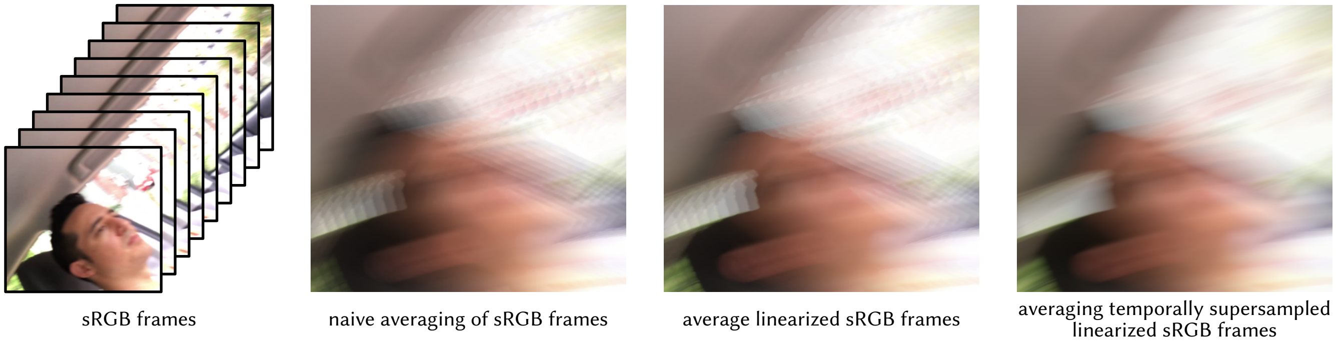

Fig. S2. Three methods for generating blurry images from a video of sRGB images. The input video frames are shown on the left. Observe that our linear interpolation approach avoids artifacts due to temporal discretization while also adhering to the blur integral of Equation 1 in the main paper.

Then, following Vaswani et al. [2017] and Mildenhall et al [2021], we define the sinusoidal positional encoding function to be

$$\gamma(t) = [\cos(2\pi\nu_1 t), \sin(2\pi\nu_1 t), \ldots, \cos(2\pi\nu_N t), \sin(2\pi\nu_N t)], \tag{S4}$$

where ν_i is an encoding frequency and N is the number of such frequencies. Lastly, we encode the time interval of the latent input motion-blurred image in a similar fashion, after first replicating $\mathcal{T}$ four times to create a vector $\tilde{\mathcal{T}} = [\mathcal{T}, \mathcal{T}, \mathcal{T}, \mathcal{T}]$ that matches the dimension of $\tilde{\mathcal{T}_i}$.

Padding. Our training pipeline requires a variable number of frames to be predicted. Thus, our model requires padding when the input frames are less than 16. Thus, we replicate the last individual frame and time encoding to match our prediction size of 16 frames. For example, in the case where we wish to predict just two individual frames, we set $\tilde{\mathcal{T}_1} = [\mathcal{T}_1, \mathcal{T}_1, \mathcal{T}_2, \mathcal{T}_2]$ as the time encoding for the first latent frame. The remaining latent frames are given the encoding $\tilde{\mathcal{T}_i} = [\mathcal{T}_2, \mathcal{T}_2, \mathcal{T}_2, \mathcal{T}_2]$ for $i = 2, 3, 4, 5, 6$. At inference time we include this type of padding but exclude the padded frames from the final output.

S2.2 Dataset

As stated in the main paper, we fine-tune a generalized version of our model on a diverse compilation of high-FPS videos drawn from GoPRO [Nah et al. 2017] (33 clips of average duration of 4s, recorded in 240 fps), Adobe240 [Su et al. 2017] (131 clips of average duration of 4s, recorded in 240 fps), REDS [Nah et al. 2019] (120 FPS), iPhone240 [Shimizu et al. 2023] (231 clips of average duration of 1 s, recorded in 240 fps), and Sports240 [Chen and Jiang 2024] (208 clips of average duration of 0.5s, recorded in 240 fps).

Capturing data for motion deblurring is a challenging problem. Most deblurring works utilize a synthetic approach for generating motion blur. Motion blur is an integration of light over an exposure period. An approximation of this can be achieved by averaging frames over the same period from high-FPS videos to generate a synthetic blurred image. This formulation is simple and allows for many sizes of motion blur to be simulated. However, even with high-FPS videos, this discretization leads to artifacts due to undersampling and dead times between frames. Additionally, many of the works perform the averaging in the gamma sRGB space which is non-linear. Ideally, the camera response function is known so the averaging can be done in linear space.

For our simulation of motion-blurred images, we perform averaging of input frames in linear sRGB space. We note that both the GOPRO and REDS datasets implement this, but other datasets like Adobe240 do not. Performing averaging in a linear color space is more realistic for blur formation [Nah et al. 2019, 2017]. Although linear sRGB doesn't estimate the true camera response function, this is still an improvement over averaging in gamma-encoded sRGB.

Additionally, we address the undersampling problem. Although we have high-FPS videos, we notice that averaging the images results in artifacts—the blur does not look smooth, as it would in a real captured image. Thus, we run a frame interpolator to temporally upsample all videos to 1920 FPS. Then, we perform averaging using these frames to simulate motion-blurred images (see Figure S2).

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